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UNDERGROUND CABLE FAULT DETECTION

A PROJECT REPORT

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Mrs. Vaishali Rajeshirke

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**Maharashtra State Board of
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MAHARASHTRA STATE BOARD OF TECHNICAL EDUCATION CERTIFICATE

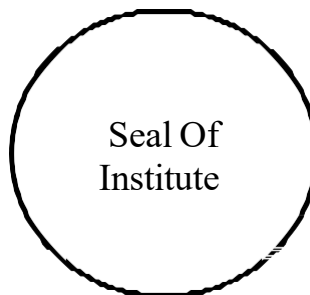
This is to certify that **Mr. Dhruv Ahire** of **Electronics & Tele-communication Engineering** From **V.E.S. Polytechnic Institute (0004)** having Enrollment no. **2200040446** has completed project of final year having title **“UNDERGROUND CABLE FAULT DETECTION ”** during **academic year 2024 – 25**. The project completed by in a group consisting of **four- person** under the guidance of faculty guide **Mrs.Vaishali Rajeshirke**

Name and Signature of Guide: **Mrs.Vaishali Rajeshirke.**

External Examiner

Head of the Department

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Certificate

Certificate

Acknowledgment

We would like to express our sincere gratitude to several individuals and organizations for supporting me throughout our Project.

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ABSTRACT

The objective of this project is to determine the distance of underground cable fault from base station in kilometers using an Arduino. The underground cable system is a common practice followed in many urban areas. While a fault occurs for some reason, at that time the repairing process related to that particular cable is difficult due to not knowing the exact location of the cable fault. The proposed system is to find the exact location of the fault.

The project uses the standard concept of Ohms law i.e., when a low DC voltage is applied at the feeder end through a series resistor (Cable lines), then current would vary depending upon the location of fault in the cable. In case there is a short circuit (Line to Ground), the voltage across series resistors changes accordingly, which is then fed to inbuilt ADC of Arduino to develop precise digital data for display in kilometres.

The project is assembled with a set of resistors representing the cable length in km and the fault creation is made by a set of switches at every known km to cross check the accuracy of the same. The fault occurring at a particular distance, the respective phase along with the distance is displayed on the LCD. The same information is also sent to the concerned authority mobile phone over GSM, interfaced to the Arduino.

Further this project can be enhanced by using capacitor in an ac circuit to measure the impedance which can even locate the open circuited cable, unlike the short circuited fault only using resistors in DC circuit as followed in the above proposed project.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

To supply electric power we use Power cables. To avoid unwanted interference, we place them underground. This makes it very tough to work out the precise location of the faults that occur. A fault might occur thanks to a many of reasons such as digging, earthquake, construction work, etc. The maintenance process associated with that specific line is difficult thanks to unknown location of the fault within the line. The motive of the project is to detect and locate the fault in underground cable. The system developed here works on the idea of Ohm's law. The proposed technique is used for identification as well as it is also used to send the information about the fault to the authority using GSM and for security of people it cut the power supply on that particular location and display the fault location in KMs on LCD Display.

The underground cable is frequently followed in many urban areas. While fault occurs for a few reasons, at that point the repairing process associated with that specific cable is difficult thanks to not knowing the precise location of cable fault. Fault in cable is represented as: • Any defect, • Inconsistency, • fragility that affects purpose of cable • Current is diverted from the intended path, From many decades, low voltage and high voltage distribution lines of underground cables are operated worldwide. To reduce the effect of distribution networks to environment underground high voltage cables are used more and more. Underground cables are widely utilized in power distribution networks thanks to the benefits of underground connection, involving more security than overhead lines in inclemency, less susceptible to damage by storms or lightning. It is less costly for shorter distance, ecofriendly and low maintenance. But if any fault occurs in cable, then it is difficult to locate fault. So, this project is employed to detect the situation of fault in digital way

. The main purpose of locating the faulty point in an underground cable in order is to ease quicker repair, improve the system stability and reduced blackout period. The underground cable system is very convenient for distribution mainly in urban areas such as, metropolitan cities, airport and defense services. The paper uses the quality concept of Ohms law. This project contains a set of resistors representing cable length in KM's to find the distance of fault in KMs and fault creation is made by a set of switches at every known KM. The fault generated at a specific distance and therefore the respective phase is displayed on a LCD interfaced to the arduino

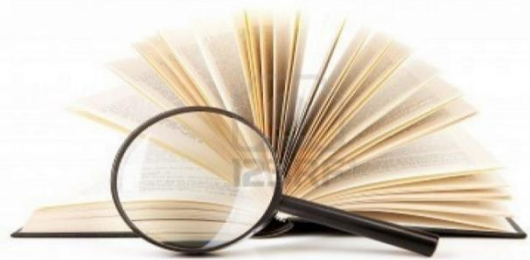
CHAPTER 2

LITERATURE SURVEY

2.1 LITERATURE SURVEY

1. The Need for Underground Cable Detectors

The rapid expansion of urban infrastructure has led to an increased reliance on underground cable systems to distribute essential utilities such as electricity, telecommunications, and data services. While these cables are better protected from environmental hazards compared to their overhead counterparts, they are not immune to faults. Detecting faults in underground cables is more complicated than in overhead systems, as the latter allows for visual inspections. The hidden nature of underground cables presents unique challenges when faults occur, making early detection vital to prevent service interruptions and ensure public safety.



Literature Review

Underground cables are subject to various risks, such as physical damage caused by construction activities, natural wear and tear due to aging, moisture ingress, and thermal stress. These factors can lead to insulation failure, short circuits, and other issues that interrupt the service. The consequences of undetected or improperly managed cable faults can be severe, including power outages, communication blackouts, and increased operational costs for utility companies. Moreover, some faults may lead to safety hazards like electrical fires or electrocution.

Traditional methods of underground cable fault detection, such as Time-Domain Reflectometry (TDR), Frequency-Domain Reflectometry (FDR), and ground-penetrating radar (GPR), are effective but expensive, labor-intensive, and often require highly specialized equipment. These methods typically involve sending electrical pulses through the cables and analyzing the reflected signals to detect and locate faults. However, such technologies are not always accessible or cost-effective for smaller utility companies or for large-scale infrastructures like underground distribution networks. Thus, there is a pressing need for more affordable and efficient solutions for underground cable fault detection.

Arduino-based systems for fault detection represent a significant step forward in

addressing this need. Arduino, an open-source electronics platform known for its affordability and ease of use, has gained popularity as a tool for engineers and hobbyists alike. Using Arduino for underground cable fault detection offers a more intensive, and often require highly specialized equipment. These methods typically involve sending electrical pulses through the cables and analyzing the reflected signals to detect and locate faults. However, such technologies are not always accessible or cost-effective for smaller utility companies or for large-scale infrastructures like underground distribution networks. Thus, there is a pressing need for more affordable and efficient solutions for underground cable fault detection.

Arduino-based systems for fault detection represent a significant step forward in addressing this need. Arduino, an open-source electronics platform known for its affordability and ease of use, has gained popularity as a tool for engineers and hobbyists alike. Using Arduino for underground cable fault detection offers a more cost-effective and accessible alternative to traditional detection systems, especially for smaller companies or regions with limited resources. Arduino-based systems allow for real-time monitoring of cable parameters, such as voltage, current, and resistance, enabling early detection of abnormalities that indicate a fault. These systems can be customized and scaled according to the complexity of the network and the specific types of faults they are designed to detect.

One of the key features of an Arduino-based underground cable fault detection system is its ability to measure the resistance of the cable at different points, allowing the system to pinpoint the location of the fault with reasonable accuracy. This method is particularly effective for identifying common faults like short circuits, open circuits, and insulation failures. In the event of a fault, the system can alert the operator through a user interface or a remote monitoring system, allowing for quick intervention to repair the damage and restore service. The development of these systems marks a shift toward more accessible and user-friendly solutions for maintaining underground cable networks.

2. History of Underground Cable Fault Detection Systems?

The concept of underground cable fault detection has evolved over the decades, with many contributors playing a role in its development. The initial foundation for underground cable fault detection can be traced back to the early 20th century, when engineers began using basic electrical principles, such as resistance and capacitance measurements, to locate faults in underground power cables. Early methods were rudimentary but laid the groundwork for more advanced technologies.

One of the earliest pioneers in cable fault detection was Dr. Oliver Heaviside, a self-taught British electrical engineer and physicist, who made significant contributions to the understanding of electrical circuits and transmission lines in the late 19th century. Heaviside's work on the behavior of electrical signals in cables led to the development of techniques that could be used to detect faults based on changes in the signal's behavior as it traveled along the cable. His insights into transmission line theory were later used to develop more sophisticated methods for detecting faults in both overhead and underground cables.

In modern times, the development of underground cable fault detection systems has been driven by advances in electronics, signal processing, and embedded systems. Several companies and research institutions have contributed to the refinement of these technologies, with a focus on improving accuracy, reducing costs, and making the systems more accessible. For example, companies like Megger and AVO International have been at the forefront of developing high-end cable testing and fault detection equipment. These devices typically use techniques such as TDR, FDR, and high-voltage testing to detect and locate faults in underground cables. However, the use of Arduino for underground cable fault detection is a more recent innovation, reflecting the broader trend toward open-source, affordable, and DIY electronics solutions. The Arduino platform itself was founded by a group of engineers and designers, including Massimo Banzi and David Cuartielles, in 2005.

Their goal was to create an easy-to-use microcontroller platform that could be used for a wide range of applications, from hobbyist projects to industrial systems. The versatility and affordability of Arduino have made it an attractive option for developing fault detection systems, especially for educational institutions, small businesses, and low-budget projects.

Arduino's role in underground cable fault detection is still in its early stages, but the community of developers and engineers using the platform continues to grow. Online forums, open-source projects, and academic papers have documented numerous applications of Arduino in fault detection, showcasing the platform's potential to revolutionize the field.

2.2 DEFINITION OF PROBLEM STATEMENT

As urban development and infrastructure projects expand, the risk of damaging underground utilities, such as power cables and communication lines, grows significantly. Conventional methods of locating these cables, such as manual digging and reliance on outdated blueprints, are not only inefficient but also increase the chances of accidental damage. This can result in costly repairs, service interruptions, and even serious safety hazards for workers.

This project aims to develop an Arduino-based underground cable detector prototype, addressing the need for a safer, more reliable, and cost-effective solution for locating buried cables. By employing sensors, signal processing, and real-time data feedback, this project seeks to demonstrate a scalable system that minimizes excavation risks and enhances safety for construction and maintenance teams. This prototype offers a potential solution for efficient underground cable detection that could be applied in various fields, from urban planning to utility management.

2.3 COMPONENTS DESCRIPTION

1. Arduino UNO

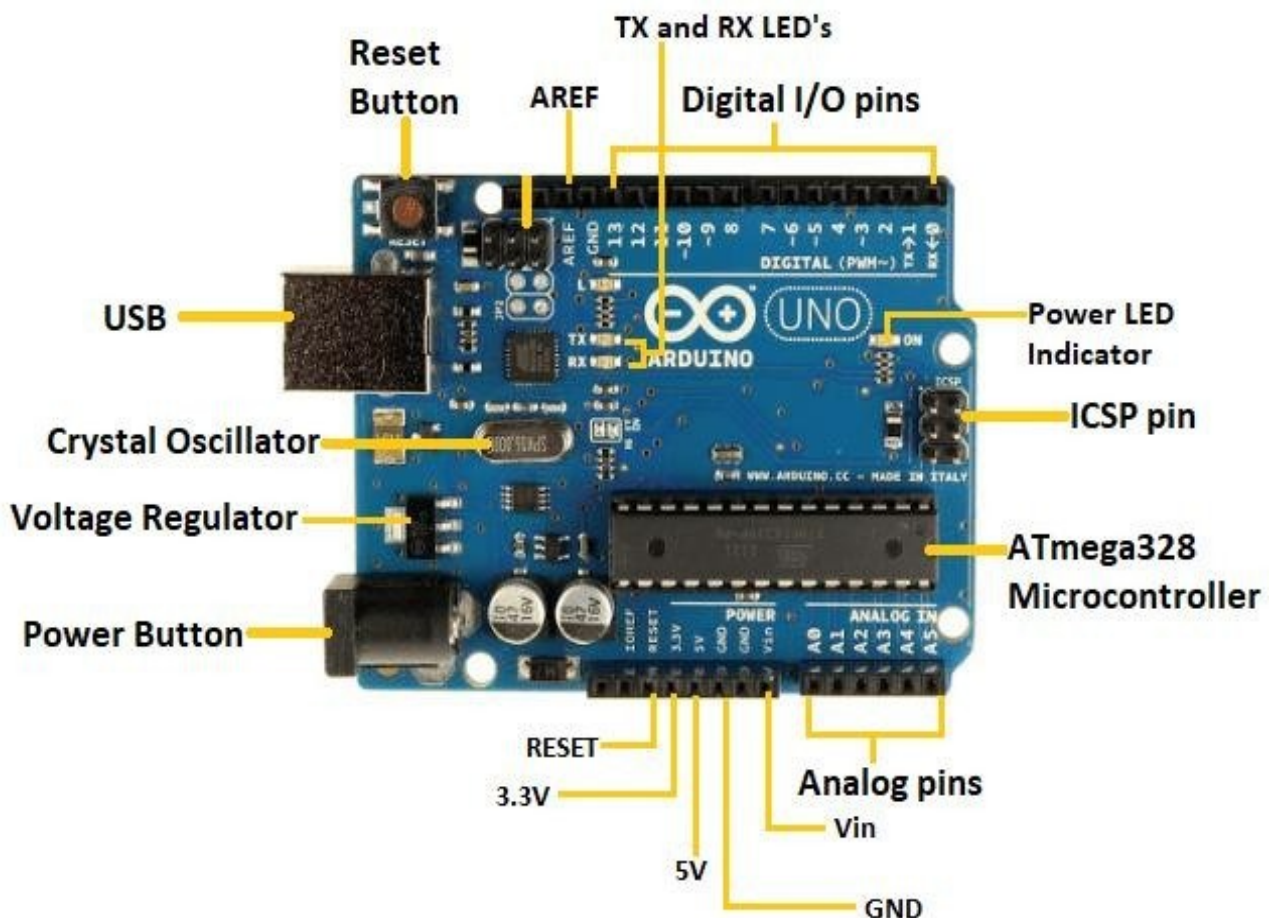


Figure 2.1 Arduino UNO

Arduino UNO is based on an Atmega328P microcontroller. It is easy to use compared to other boards, such as the Arduino Mega board, etc. The board consists of digital and analog Input/Output pins (I/O), shields, and other circuits. The Arduino UNO includes 6 analog pin inputs, 14 digital pins, a USB connector, a power jack, and an ICSP (In-Circuit Serial Programming) header. It is programmed based on IDE, which stands for Integrated Development Environment. It can run on both online and offline platforms. The **Arduino Uno** is a microcontroller board based on the Atmega328. Arduino is an open-source, prototyping platform and its simplicity makes it ideal for hobbyists to use as well as professionals.

Features

- Microcontroller: Atmega328.
- Operating Voltage: 5V.
- Input Voltage (limits): 6-20V.
- Digital I/O Pins: 14 (of which 6 provide PWM output)
- Analog Input Pins: 6.
- DC Current per I/O Pin: 40 Ma.
- DC Current for 3.3V Pin: 50 Ma.
- Flash Memory: 32 KB of which 0.5 KB is used the by bootloader
- SRAM: 2 KB (Atmega328)
- EEPROM: 1 KB (Atmega328)
- Clock Speed: 16 MHz



2. 16 x 2 LCD Display

Figure 2.2 16 x 2 LCD Display

A 16 x 2 LCD Display is used to display the Message about the Faults, their Distance and Latitude and Longitude Coordinates. In addition to this, it can be used to know what Process is going on in the Microcontroller from Initialization of the devices. Its working voltage is 5V. It is interfaced with Arduino in 4-bit Operating Mode.

Features

- Operating Voltage: 4.7V to 5.3V.
- Operating Current 1mA (without backlight)
- Can display (16x2) 32 Alphanumeric Characters.
- Custom Characters Support.
- Works in both 8-bit and 4-bit Mode

3 GSM 800A

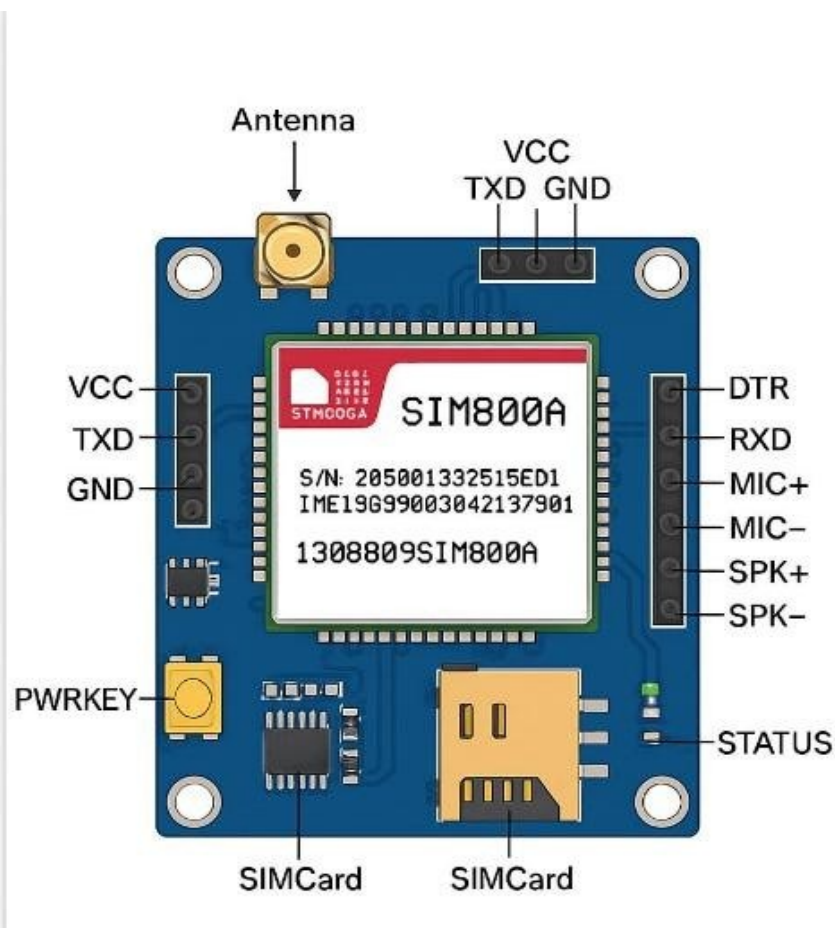


Figure 2.3 GSM Module (GSM 800A)

The SIM800A is a GSM/GPRS module that enables communication via mobile networks. It is typically used in embedded systems to provide wireless connectivity for applications such as remote monitoring, IoT devices, and GSM-based communication solutions.

Features

- **GSM/GPRS Support:** Allows sending SMS, making voice calls, and accessing the internet via GPRS.
- **AT Commands:** Controlled through AT commands, which are easy to implement and integrate into microcontroller-based projects.
- **Compact Size:** Small form factor, making it suitable for embedded and portable applications.
- **Multiple Frequency Bands:** Works on 850/900/1800/1900 MHz frequencies, allowing global coverage.
- **Low Power:** Energy-efficient design, ideal for battery-operated systems.
- **External Antenna:** Option for connecting an external antenna for improved signal reception.

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Each relay on the module is a Single Pole Double Throw (SPDT) type, meaning it has one common terminal and two different contact terminals that can be connected in various configurations. In this project, an array of SPDT relays is utilized to indicate faults at different distances. The 3-channel setup allows for the monitoring and control of multiple points or devices simultaneously, making it suitable for more complex systems requiring multiple fault indicators.

Features:

- **Electromagnetic Operation:** Uses an electromagnetic coil to open/close contacts.
- **Isolation:** Provides electrical isolation between control and output circuits.
- **Control High Power with Low Power:** Switches high-voltage or high-current devices using a small control signal.
- **Multiple Contacts:** Available in different configurations (e.g., SPST, DPDT).
- **Types:** Mechanical (with moving parts) and Solid-State (no moving parts).
- **Durability:** Can handle many switching cycles, though mechanical relays wear over time.
- **Coil Voltage:** Available in various voltages (e.g., 5V, 12V, 24V).
- **Contact Types:** Normally Open (NO) and Normally Closed (NC).
- **Fast Response:** Quick switching, especially in solid-state relays.

2.4 TOTAL COSTING OF PROJECT

Sr. No	Name of the Component	Specifications	Quantity	Total Costing
1.	Arduino UNO	ATmega328p	1	700
2.	LCD Display	16*2 with I2C	1	200
3.	GSM Module	GSM 800A	1	1000
4.	Transformer	Step up	1	1800
5.	Relay Module	Single Channel	3	250
6.	Sim Card	VI Card	1	500
7.	Single strand wire	-	5m	250
8.	9 Volt Adapter	9V , 2A	1	350
Total Costing of Project				₹5050

Table No. 2.4.1

CHAPTER 3

SCOPE OF PROJECT

3.1 Future SCOPE OF PROJECT

The future scope of the **Underground Cable Fault Detection** project

includes advancements in IoT, AI, and automation for enhanced efficiency. Integration with cloud computing and AI-driven predictive maintenance will allow real-time monitoring and early fault detection. GPS and GIS-based systems can improve fault localization accuracy,

reducing downtime, while mobile app integration can provide instant fault alerts and graphical fault mapping. Future improvements may also include renewable energy-powered sensors, drone-based inspections, and blockchain for secure data transmission, making underground cable monitoring more reliable and cost-effective.



3.2 Boundaries of this project:

In this **Underground Cable Fault Detection** system, the GSM 800A module is used for fault alerts, but it operates within a certain range and at a frequency of **800 MHz**, limiting its coverage area. The system relies on a **9V power supply** for the GSM module, making it essential to maintain a stable power source. This project is designed to detect and locate faults in underground cables but does not physically repair them, requiring manual intervention for maintenance. While it enhances fault detection efficiency, **environmental factors like moisture and interference** may affect sensor accuracy. Future improvements can include IoT integration for real-time monitoring and enhanced communication using wireless technologies.

3.3 Objective of this project:

The primary objective of the **Underground Cable Fault Detection** project is to provide an efficient and reliable system for identifying faults in underground cables, reducing downtime and maintenance efforts. The system should accurately detect and locate faults, ensuring quick troubleshooting and repair. It should operate consistently over a long period without frequent failures or recalibrations. The system should be user-friendly, with a simple interface for technicians to monitor fault data and alerts. It should be able to handle multiple fault detections simultaneously without any downtime or malfunction. The GSM-based alert system should send real-time notifications to maintenance teams for immediate action. The system should be **cost-effective, easy to install, and require minimal modifications to the existing power infrastructure**. It should also be compatible with different types of underground cable networks and adaptable for future enhancements, such as IoT integration for remote monitoring and predictive maintenance.

3.4 Impact of project on the society:

The **Underground Cable Fault Detection** system can significantly improve power supply reliability by quickly identifying faults in underground cables, reducing downtime, and minimizing power outages. This enhances the efficiency of electrical distribution networks, benefiting both residential and industrial consumers. By preventing prolonged blackouts, the system contributes to economic stability and ensures uninterrupted operation of essential services such as hospitals, businesses, and communication networks. Additionally, early fault detection reduces the risk of accidents like electrical fires or shocks, enhancing public safety. The project also helps utility companies save on repair costs and labor by pinpointing faults accurately, leading to **faster maintenance and lower operational expenses**. In the long run, the system supports the development of smarter, more resilient electrical infrastructure, making energy distribution more **sustainable and efficient** for society.

3.5 Goal of the project:

The goal of the Underground Cable Fault Detection project is to provide a reliable and efficient system for identifying faults in underground power cables, ensuring a stable and uninterrupted power supply. The project aims to replace traditional manual fault detection methods with a modern and automated solution that uses electrical principles and GSM technology to accurately detect and locate faults. The primary objective is to design, develop, and implement a system that can quickly identify fault locations, reducing repair time, maintenance costs, and power outage durations. The system should be accurate, user-friendly, and reliable, providing real-time alerts to maintenance teams for immediate action. The project also aims to integrate with smart grid technologies and IoT-based monitoring systems for a more advanced and comprehensive power management solution. The system should be scalable, adaptable to different power infrastructures, and contribute to a more efficient, cost-effective, and sustainable energy distribution network.

3.6 Achievement of this project:

1. We successfully developed a **more efficient and automated** system for detecting underground cable faults, reducing reliance on manual inspection methods.
2. The system enhances **power supply reliability** by quickly identifying faults, minimizing downtime, and preventing prolonged power outages.
3. Real-time **fault location detection** enables faster response and repair, reducing maintenance costs and improving service efficiency.
4. The use of **GSM technology** ensures that fault alerts are immediately communicated to maintenance teams, leading to quicker troubleshooting.
5. The project provides a **cost-effective solution** for power companies, reducing labor and operational expenses while improving infrastructure efficiency.
6. The system is **scalable and adaptable**, allowing future integration with IoT and smart grid technologies for enhanced monitoring and predictive maintenance.

CHAPTER 4

METHODOLOGY

4.1 PROPOSED METHODOLOGY

- Advanced Methods like Fault Detection, Fault Diagnosis have Become Important for Many Technical processes for the Improvement of Reliability, Safety and Efficiency. There are Two Types of Fault location Methods namely Online and Offline methods.
- Online Method: -To determine the fault points this method utilizes & processes the sampled voltages & current. Online method for underground cable is less than overhead lines.

The online method is a real-time fault detection technique that continuously monitors and analyzes voltage and current signals in electrical cables to identify faults as they occur. It uses sensors and automated systems to process these signals and detect irregularities like short circuits, open circuits, or insulation failures.

For underground cables, the online method is less common than in overhead lines due to challenges like signal attenuation, insulation effects, and complex cable networks. However, when used, it helps in early fault detection, reducing downtime, and improving system reliability by providing immediate alerts through communication systems like GSM or IoT-based networks.

- The system continuously monitors the **underground cable network** to detect faults by analyzing electrical parameters such as **voltage and current variations**.
- The **Arduino microcontroller** processes real-time data and compares it with predefined values stored in memory to identify abnormalities.
- Once a fault is detected, the **output signal is activated**, which can be used to trigger an alert or notify maintenance teams.
- An **LCD display** is used to **show fault status**, indicating whether a fault is detected and providing details about its approximate location.
- The entire system consists of three main components:
- **Primary stage** – Detecting voltage or current fluctuations and capturing fault signals.
- **Secondary stage** – Processing the fault data and transmitting alerts via the **GSM module** to the registered mobile number.
- **Final stage** – Displaying the fault location on the **LCD screen** and sending real-time notifications, enabling quick response and repair.
- If a fault is detected, the **system alerts the maintenance team**, and if no fault is found, the system continues monitoring for any future issues.

The functionalities of this complete Underground cable detection :-

The system continuously monitors the underground cable network to detect faults using voltage and current analysis.

The **Arduino microcontroller** processes real-time electrical signals and compares them with predefined fault conditions stored in memory.

When a fault is detected, the system triggers an alert and activates the **GSM module** to send notifications to the maintenance team.

An **LCD display** is used to show fault status, including the fault location and type, ensuring easy monitoring.

The system operates in three key stages:

1. **Fault Detection:** The system continuously monitors electrical parameters and identifies abnormal conditions.
2. **Fault Location:** It calculates and displays the approximate fault distance, helping technicians locate and repair it quickly.
3. **Fault Notification:** The detected fault information is sent via **GSM-based SMS alerts** to maintenance personnel for immediate action.

If a fault is found, the **LCD display** shows fault details, enabling technicians to take corrective measures and restore power efficiently.

4.2 CHALLENGES FACED & ENCOUNTER

While Developing an underground cable detection using Arduino uno and GSM 800A module we have faced many problem while doing this project. So we have to manage to encounter problem by step by step manner. We have faced many problems as following below

We have faced lots of challenges in Different part:

- GSM PART
- LCD PART
- CODING ISSUES

➤ GSM PART:

Network coverage: One of the major challenges with the GSM part of the system is network coverage. If the area where the system is installed has poor network coverage, the system may not function properly, and users may experience delays or inability to access the lock.

SIM card issues: The system relies on a SIM card to establish communication between the device and the server. Any issues with the SIM card, such as expired or faulty SIM card, can prevent the system from functioning properly.

SMS costs: Another challenge with the GSM part of the system is the cost of SMS. Since the system relies on SMS, the cost of SMS can be significant, especially if the system is used frequently.

Security issues: Security is a significant concern with any GSM-based system. If the system is not properly secured, hackers can exploit vulnerabilities in the system

➤ LCD PART:

Display quality: The quality of the display is a significant challenge. The display must be clear and easy to read for the user to relevant information.

Contrast adjustment: The contrast of the display must be properly adjusted to ensure that the text is readable. Failure to adjust the contrast may cause the text to be too dim or too bright, which may be difficult to read.

Size of the display: The size of the display is another important factor to consider. If the display is too small, users may find it difficult to read, which can lead to errors and delays.

Connection issues: The LCD module must be properly connected to the system to function correctly. Any loose connections can cause the display to malfunction or stop working altogether.

Compatibility: Compatibility is another challenge with the LCD part of the system.

The LCD module must be compatible with the microcontroller used in the system, and any compatibility issues can lead to malfunctioning or improper functioning of the system.

➤ **CODING ISSUES:**

Complexity: The coding part of the system can be complex, and it requires a good understanding of programming concepts. Writing the code for the system requires a high level of expertise in programming, and any errors in the code can cause the system to malfunction or stop working altogether.

Debugging issues: Debugging the code is another significant challenge. Finding the source of errors in the code can be difficult, and it requires a good understanding of programming concepts.

Compatibility issues: Compatibility is another challenge with the coding part of the system. The code must be compatible with the microcontroller, the GSM module, and the LCD module used in the system. Any compatibility issues can cause the system to malfunction or not function at all.

Security issues: Security is a significant concern with any coding-based system. The system must be properly secured.

Testing issues: Testing the code is another significant challenge. The system must be thoroughly tested to ensure that it functions correctly under different conditions and scenarios.

CHAPTER 5

Details of Design, Working and Processes

5.1 BLOCK DIAGRAM

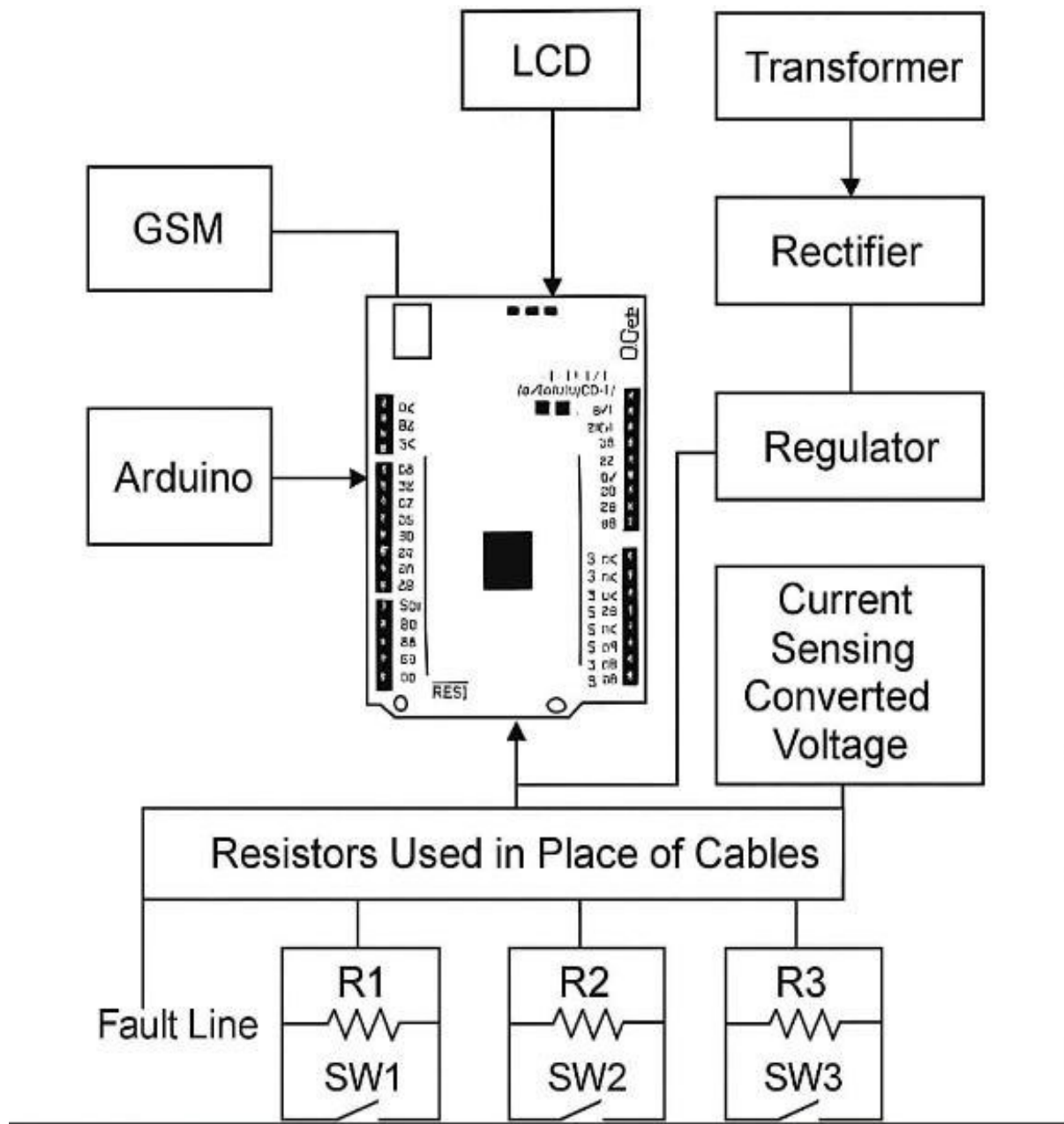
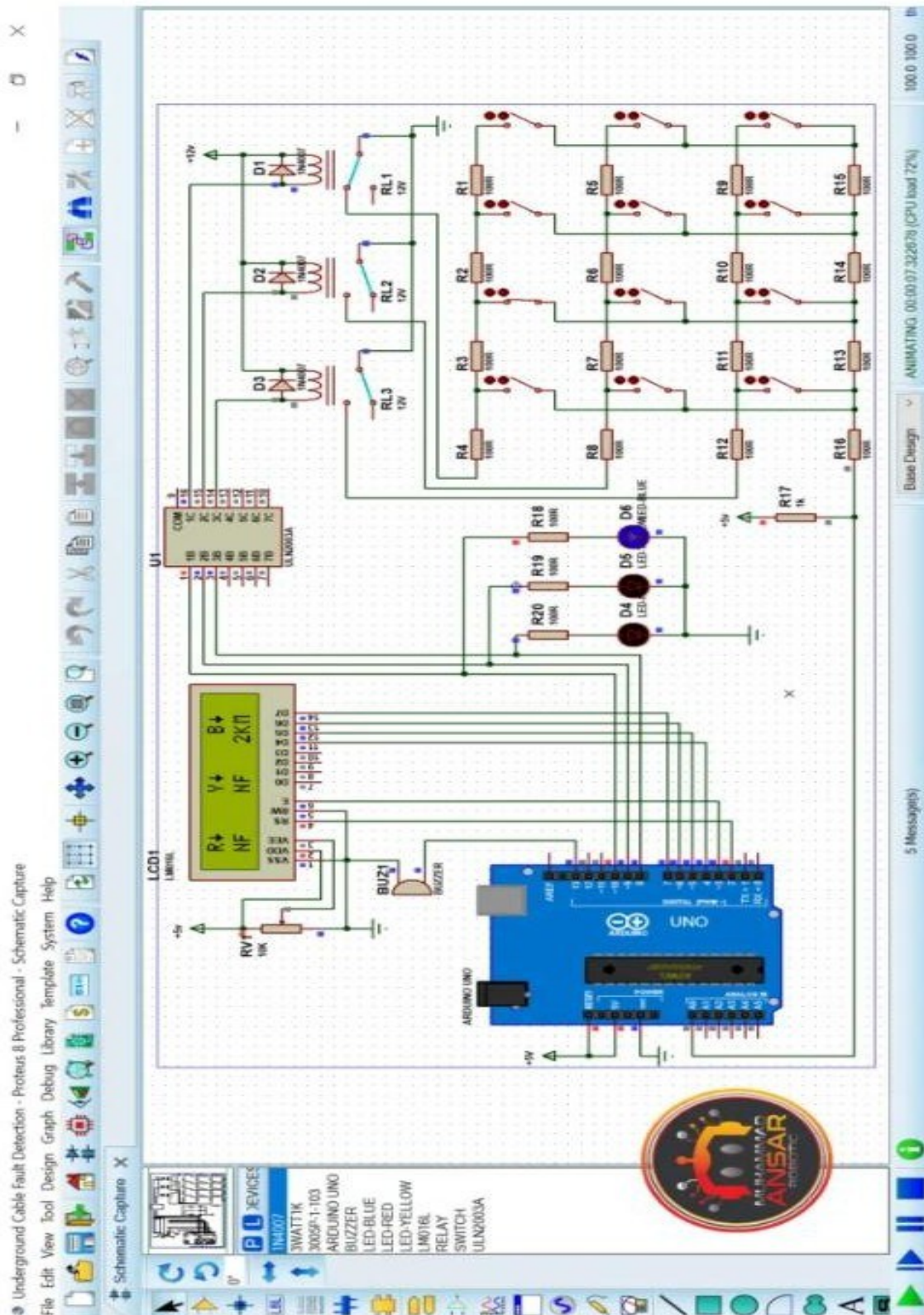


Figure 5.1: Block Diagram of underground cable fault detection

The overall working of the system is as follows:-

1. AC Supply: Provides the initial power input to the system. Typically 230V AC from the mains.
2. Transformer: Steps down the high AC voltage (230V) to a lower AC voltage (e.g., 12V AC) suitable for electronics.
3. Rectifier: Converts the stepped-down AC voltage into DC voltage using a bridge rectifier (diode-based).
4. Regulator: Regulates the DC voltage to a stable 5V or 12V DC as required by the Arduino and other modules.
5. Arduino: Acts as the main controller of the project. Reads voltage levels from the cable lines (simulated using resistors and switches). Determines if a fault has occurred and calculates the location of the fault. Sends this data to both the LCD display and the GSM module.
6. LCD Display: Shows real-time messages like "No Fault", or "Fault at 5 KM" based on which line is shorted or open. Useful for local monitoring.
7. GSM Module: Sends an SMS alert to a predefined mobile number. Message includes information like which line is faulty and distance of the fault.
8. Current Sensing → Voltage Converter: Measures the current flowing through the simulated cables. Converts it into voltage (analog) readable by Arduino's ADC (Analog to Digital Converter). Low or no current indicates a fault in that segment.
9. Resistors Used in Place of Cables: These simulate the resistance of real underground cables. Each resistor segment (R1, R2, R3) represents a certain length of cable (e.g., 5 km each). Connected with switches (SW1, SW2, SW3) to simulate faults like open circuit or short circuit.
10. Fault Switches: Manually triggered to simulate a fault condition at a specific point. When switched, the corresponding resistor path breaks or shorts, altering the current/voltage. Arduino detects this change and identifies the fault location.

5.2 INTERFACING DIAGRAM OF THE PROJECT



Underground Cable Fault Detection Proteus Simulation

Figure 5.2: Interfacing Diagram of underground cable fault detection

Diagram Explanation:-

- **Power Supply**

The relay module's **VCC** and **GND** pins are connected to the 5V and GND of the microcontroller.

- **Signal Inputs**

IN1, IN2, IN3 of the relay module are connected to **digital pins** on the microcontroller (e.g., D2, D3, D4). These pins send HIGH/LOW signals to control each relay.

- **Relay Contacts:**

Each relay (SPDT) has:

COM: Common terminal

NO: Normally open

NC: Normally closed

You connect the cable segments (or dummy loads) between these terminals to simulate working/fault conditions. For instance: When a fault is detected in a segment, the microcontroller activates the corresponding relay (sends HIGH signal). The relay switches from **NC** to **NO**, triggering a **buzzer or LED** to indicate the fault at that particular segment.

- **Logic Behind Fault Detection:**

The underground cable is divided into sections. A microcontroller sends test pulses or measures resistance/voltage drop across the segments. Based on the return signal or value, it detects which segment is faulty. Then, it activates the respective relay to indicate that fault.

5.3 Interfacing of GSM Module with Arduino

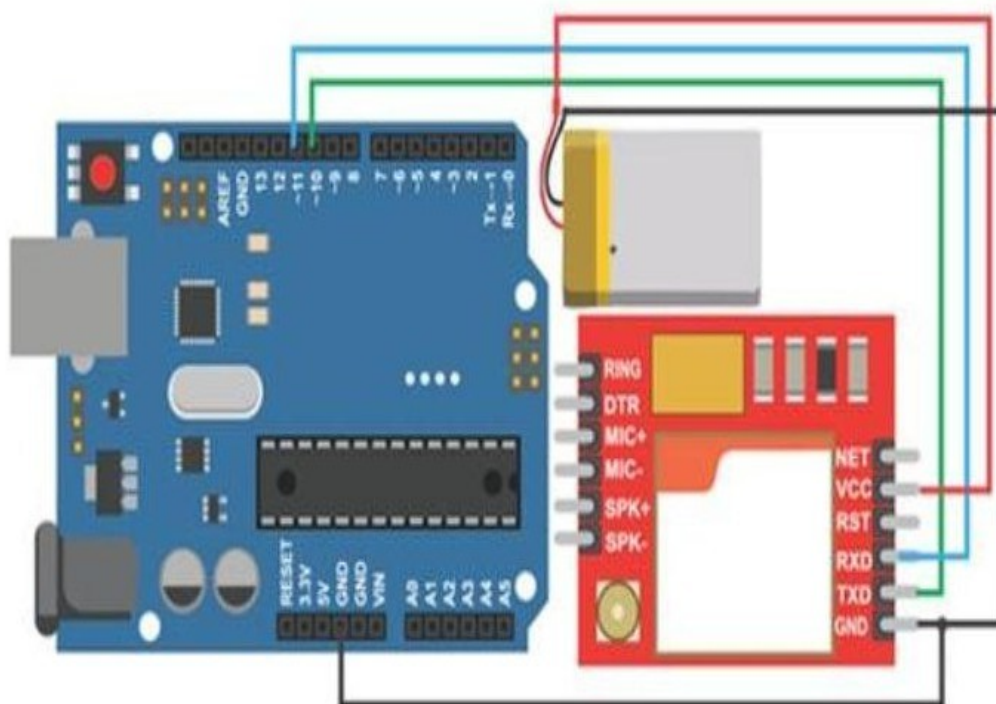
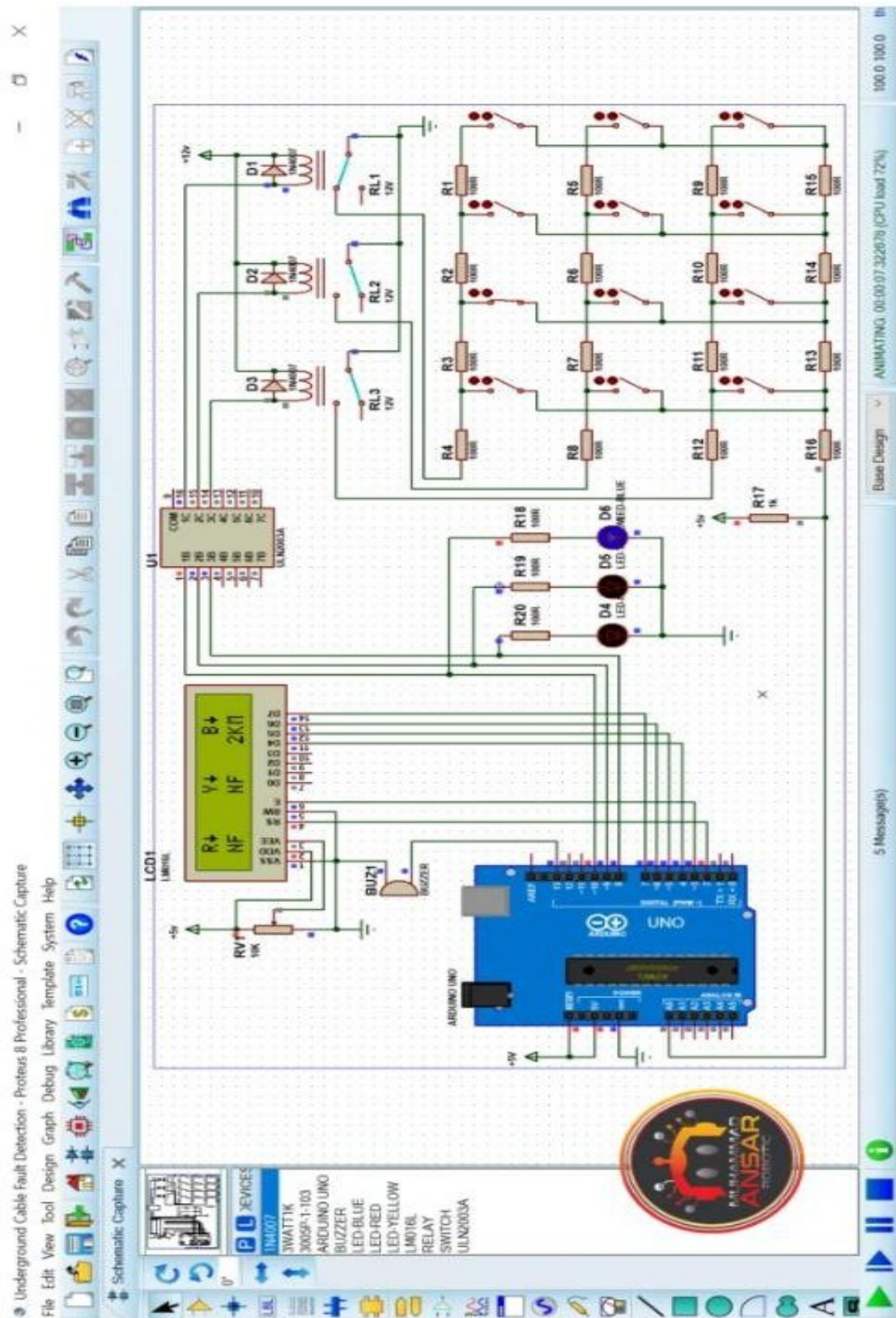


Figure 5.3: Interfacing GSM Module with Arduino

GSM Module Pin	Arduino Pin	Description
VCC	External 4V (not from Arduino)	GSM needs 3.7–4.2V
GND	GND	Common ground
TX	D3	GSM Transmit → Arduino Receive
RX	D2	GSM Receive ← Arduino Transmit

The **GSM module** is used to send/receive SMS (e.g., alert when a fault is detected). The **relay module** is controlled by Arduino based on sensor input or logic (e.g., turn on a buzzer when a fault is found). The Arduino reads data or sensor input, activates the **relay**, and also sends an **SMS alert** using the GSM module.

5.4 Circuit Diagram Designed in IDE Software (PROTEUS) :



Underground Cable Fault Detection Proteus Simulation

Figure 5.4: Design of Circuit Diagram

5.5 LCD & Arduino Interface

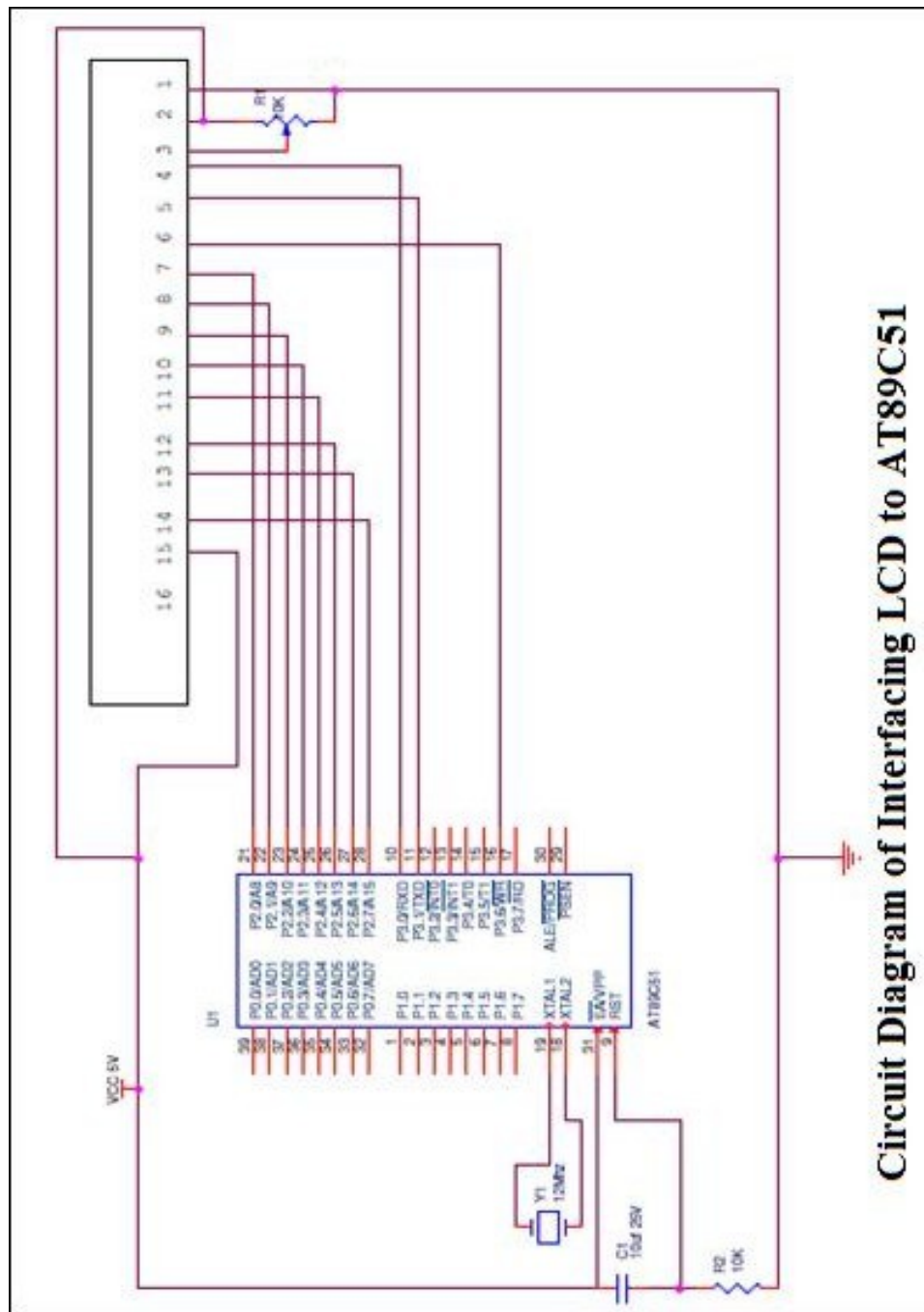


Figure 5.5: Interfacing LCD with Arduino

Pin Description:

PinNo	Function	Name
1	Ground (0V)	Ground
2	Supply voltage; 5V (4.7V – 5.3V)	V _{cc}
3	Contrast adjustment; through a variable resistor	V _{ee}
4	Selects command register when low; and data register when high	Register Select
5	Low to write to the register; High to read from the register	Read/write
6	Sends data to data pins when a high to low pulse is given	Enable
7	8-bit data pins	DB0
8		DB1
9		DB2
10		DB3
11		DB4
12		DB5
13		DB6
14		DB7
15	Backlight V _{CC} (5V)	Led+
16	Backlight Ground (0V)	Led-

16×2 LCD module commands.

16×2 LCD module has a set of preset command instructions. Each command will make the module to do a particular task. The commonly used commands and their function are given in the table below.

Command	Function
0F	LCD ON, Cursor ON, Cursor blinking ON
01	Clear screen
02	Return home
04	Decrement cursor
06	Increment cursor
0E	Display ON ,Cursor blinking OFF
80	Force cursor to the beginning of 1 st line
C0	Force cursor to the beginning of 2 nd line
38	Use 2 lines and 5×7 matrix
83	Cursor line 1 position 3
3C	Activate second line
08	Display OFF, Cursor OFF
C1	Jump to second line, position1
OC	Display ON, Cursor OFF
C1	Jump to second line, position1
C2	Jump to second line, position2

Step 1: Wiring

First, connect the pins of the LCD module to the Arduino board. Connect the following pins of the LCD module to the corresponding pins on the Arduino board:

Step 2: Upload the Library

Next, download and install the Liquid Crystal library for Arduino. This library makes it easy to control the LCD display using the Arduino board. Open the Arduino IDE, go to Sketch > Include Library > Manage Libraries, search for Liquid Crystal, and click install.

Step 3: Write the Code of Underground Cable Fault Detection**5.6 OPERATING SYSTEM (SOFTWARE)****IDE (Integrated Development Environment) Software**

The Arduino Integrated Development Environment - or Arduino Software (IDE) - contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus. It connects to the Arduino hardware to upload programs and communicate with them. The Arduino Software (IDE) makes it easy to write code and upload it to the board offline. We recommend it for users with poor or no internet connection. This software can be used with any Arduino board.

There are currently two versions of the Arduino IDE, one is the IDE 1.x.x and the other is IDE 2.x. The IDE 2.x is new major release that is faster and even more powerful to the IDE 1.x.x. In addition to a more modern editor and a more responsive interface it includes advanced features to help users with their coding and debugging.

Features of IDE 2.0

- Modern, fully featured development environment
- Dual Mode, Classic Mode (identical to the Classic Arduino IDE) and Pro Mode (File System view)
- New Board Manager
- New Library Manager
- Board List

- Basic Auto-Completion (Arm targets only)
- Git Integration
- Serial Monitor

5.7 FLOWCHART

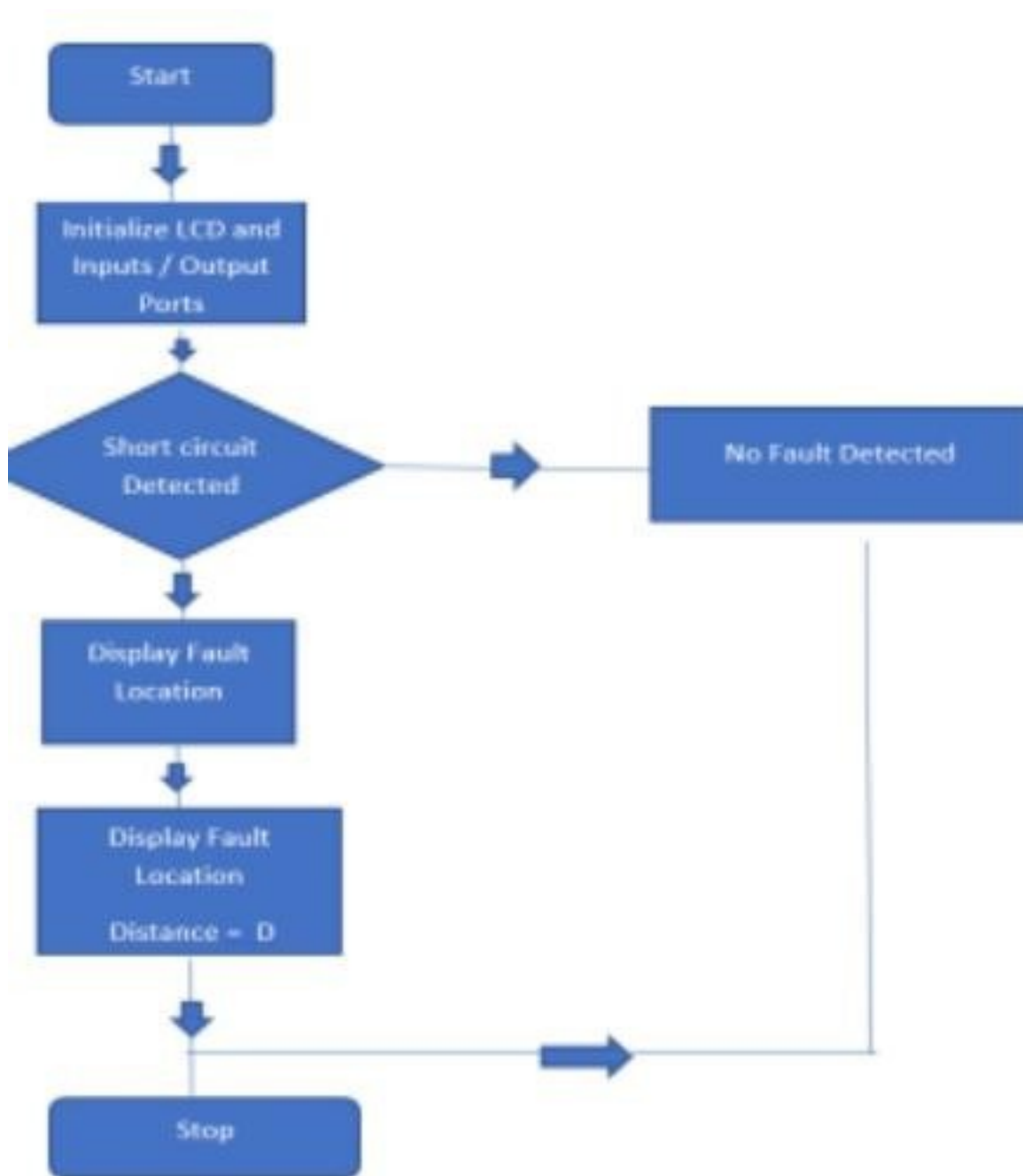


Figure 5.7: Flowchart

5.8 SOURCE CODE

Initializing Libraries and Defining Constants

1. Cable (Resistance)
2. LCD Display Section
3. GSM Section

```
#include <LiquidCrystal.h>
LiquidCrystal lcd(3, 4, 5, 6, 7, 8); //RS,EN,D4,D5,D6,D7
const int analogInPin0 = A5; // Analog input pin that the potentiometer is attached to
const int analogInPin1 = A3;
const int analogInPin2 = A0;
char num[] = "9082468683";

int RED2 = 9;
int BLUE2 = 11;
int YELLOW2 = 10;

int RR=0,YY=0,BB=0;

int REDR = 0;
int BLUEB = 0;
int YELLOWY=0;

int a1 = 0, b1 = 0, c1 = 0, d1 = 0, e1 = 0, f1 = 0, g1 = 0, h1 = 0,
i1 = 0, j1 = 0, k1 = 0, l1 = 0, m1 = 0, n1 = 0, o1 = 0, p1 = 0, q1 = 0, r1 = 0;

void setup()
{
    lcd.begin(16, 2);
    Serial.begin(9600);
    pinMode(RED2, OUTPUT);
    pinMode(BLUE2, OUTPUT);
    pinMode(YELLOW2, OUTPUT);
    digitalWrite (RED2,HIGH);
    digitalWrite (BLUE2,HIGH);
    digitalWrite (YELLOW2,HIGH);
    lcd.setCursor(0, 0); // top left
    lcd.print("UNDERGROUND CABLE ");
    lcd.setCursor(0, 1);
    lcd.print("FAULT OVER GSM ");
    delay(3000);
    lcd.clear();
    Serial.print("AT+CMGF=1 \r"); // set SMS mode to text
    delay(100);
    Serial.print("AT+CNMI=2,2,0,0,0 \r");
    delay(100);
```

```

    lcd.setCursor(0, 0); // top left
    lcd.print("R=0 Km");
    lcd.setCursor(9, 0); // top left
    lcd.print("Y=0 Km");
    lcd.setCursor(0, 1); // top left
    lcd.print("B=0 Km");
  }
void loop()
{
  while(1)
  {
    REDR = analogRead(analogInPin0);
    YELLOWY = analogRead(analogInPin1);
    BLUEB = analogRead(analogInPin2);
    //Serial.println( REDR);
    RED1();

    YELLOW1();

    BLUE1();

  }
}

void RED1()
{

  if(RR==0)
  {
    REDR = analogRead(analogInPin0);
    if(REDR>=850 && REDR<=856 )
    {
      if(a1 ==0)
      {
        Serial.print("AT\r");
        delay(200);
        Serial.print("AT+CMGF=1\r");
        delay(200);
        Serial.print("AT+CNMI=2,2,0,0,0\r");
        delay(200);
        Serial.print("AT+CMGS=");
        delay(200);
        Serial.print("");
        delay(200);
        Serial.print(num);
        delay(200);
        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("R=30Km");
        delay(200);

```

```

        Serial.write(0x1A);
        delay(200);

digitalWrite (RED2,LOW);
// Serial.println("R=30Km");
lcd.setCursor(0, 0); // top left
lcd.print("R=30Km");
a1 = 1;
RR=1;
}
}
else if(REDR>=820 && REDR<=825 )
{
    if(b1 == 0)
    {
        Serial.print("AT\r");
        delay(200);
        Serial.print("AT+CMGF=1\r");
        delay(200);
        Serial.print("AT+CNMI=2,2,0,0,0\r");
        delay(200);
        Serial.print("AT+CMGS=");
        delay(200);
        Serial.print("");
        delay(200);
        Serial.print(num);
        delay(200);
        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("R=25Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);

digitalWrite (RED2,LOW);
// Serial.println("R=25Km");
lcd.setCursor(0, 0); // top left
lcd.print("R=25Km");
b1 = 1;
RR=1;
}
}
else
if(REDR>=765 && REDR<=770 )
{
    if(c1 == 0)
    {
        //Serial.println("R=20Km");
        lcd.setCursor(0, 0); // top left
        lcd.print("R=20Km");

```

```

digitalWrite (RED2,LOW);

    Serial.print("AT\r");
    delay(200);
    Serial.print("AT+CMGF=1\r");
    delay(200);
    Serial.print("AT+CNMI=2,2,0,0,0\r");
    delay(200);
    Serial.print("AT+CMGS=");
    delay(200);
    Serial.print("");
    delay(200);
    Serial.print(num);
    delay(200);
    Serial.print("");
    delay(200);
    Serial.write(0x0D);
    delay(200);
    Serial.println("R=20Km");
    delay(200);
    Serial.write(0x1A);
    delay(200);
    c1 = 1;
    RR=1;
}
}
else
if(REDR>=670 && REDR<=695 )
{
// Serial.println("R=15Km");
if(d1 == 0)
{
    lcd.setCursor(0, 0); // top left
    lcd.print("R=15Km");
    digitalWrite (RED2,LOW);
    Serial.print("AT\r");
    delay(200);
    Serial.print("AT+CMGF=1\r");
    delay(200);
    Serial.print("AT+CNMI=2,2,0,0,0\r");
    delay(200);
    Serial.print("AT+CMGS=");
    delay(200);
    Serial.print("");
    delay(200);
    Serial.print(num);
    delay(200);
    Serial.print("");
    delay(200);
    Serial.write(0x0D);
    delay(200);
    Serial.println("R=15Km");

```

```

        delay(200);
        Serial.write(0x1A);
        delay(200);
        d1=1;
        RR=1;
    }
}
else
if(REDR>=510 && REDR<=515 )
{
    //Serial.println("R=10Km");
    if(e1== 0)
    {
        lcd.setCursor(0, 0); // top left
        lcd.print("R=10Km");
        digitalWrite (RED2,LOW);
        Serial.print("AT\r");
        delay(200);
        Serial.print("AT+CMGF=1\r");
        delay(200);
        Serial.print("AT+CNMI=2,2,0,0,0\r");
        delay(200);
        Serial.print("AT+CMGS=");
        delay(200);
        Serial.print("");
        delay(200);
        Serial.print(num);
        delay(200);
        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("R=10Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);
        e1 = 1;
        RR=1;
    }
}
else
if( REDR<=400 )
{
    // Serial.println("R=5Km");
    if(f1 == 0)
    {

        lcd.setCursor(0, 0); // top left
        lcd.print("R=5Km");
        digitalWrite (RED2,LOW);
        Serial.print("AT\r");
        delay(200);

```



```

        Serial.print("AT+CMGF=1\r");
        delay(200);
        Serial.print("AT+CNMI=2,2,0,0,0\r");
        delay(200);
        Serial.print("AT+CMGS=");
        delay(200);
        Serial.print("");
        delay(200);
        Serial.print(num);
        delay(200);
        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("R=5Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);
    f1 =1;
    RR=1;
}
}
}
}

void YELLOW1()
{

    if(Y==0)
    {
        YELLOW = analogRead(analogInPin1);
        if(YELLOW>=850 && YELLOW<=856 )
        {
            if(g1 == 0)
            {

                lcd.setCursor(9, 0); // top left
                lcd.print("Y=30Km");
                digitalWrite (YELLOW2,LOW);
                Serial.print("AT\r");
                delay(200);
                Serial.print("AT+CMGF=1\r");
                delay(200);
                Serial.print("AT+CNMI=2,2,0,0,0\r");
                delay(200);
                Serial.print("AT+CMGS=");
                delay(200);
                Serial.print("");
                delay(200);
                Serial.print(num);
                delay(200);
            }
        }
    }
}

```

```

        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("Y=30Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);
        g1 = 1;
        YY=1;
    }
}
else if(YELLOWY>=820 && YELLOWY<=825 )
{

    if(h1 == 0)
    {
        lcd.setCursor(9, 0); // top left
        lcd.print("Y=25Km");
        digitalWrite (YELLOW2,LOW);
        Serial.print("AT\r");
        delay(200);
        Serial.print("AT+CMGF=1\r");
        delay(200);
        Serial.print("AT+CNMI=2,2,0,0,0\r");
        delay(200);
        Serial.print("AT+CMGS=");
        delay(200);
        Serial.print("");
        delay(200);
        Serial.print(num);
        delay(200);
        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("Y=25Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);
        h1 = 1;
        YY=1;
    }
}
else
if(YELLOWY>=765 && YELLOWY<=770 )
{

    if(i1 == 0)
    {
        lcd.setCursor(9, 0); // top left
        lcd.print("Y=20Km");

```

```

digitalWrite (YELLOW2,LOW);
  Serial.print("AT\r");
  delay(200);
  Serial.print("AT+CMGF=1\r");
  delay(200);
  Serial.print("AT+CNMI=2,2,0,0,0\r");
  delay(200);
  Serial.print("AT+CMGS=");
  delay(200);
  Serial.print("");
  delay(200);
  Serial.print(num);
  delay(200);
  Serial.print("");
  delay(200);
  Serial.write(0x0D);
  delay(200);
  Serial.println("Y=20Km");
  delay(200);
  Serial.write(0x1A);
  delay(200);
  i1 =1;
  YY=1;
}
}
else
if(YELLOWY>=670 && YELLOWY<=695 )
{

if(j1 == 0);
{
  lcd.setCursor(9, 0); // top left
  lcd.print("Y=15Km");
  digitalWrite (YELLOW2,LOW);
  Serial.print("AT\r");
  delay(200);
  Serial.print("AT+CMGF=1\r");
  delay(200);
  Serial.print("AT+CNMI=2,2,0,0,0\r");
  delay(200);
  Serial.print("AT+CMGS=");
  delay(200);
  Serial.print("");
  delay(200);
  Serial.print(num);
  delay(200);
  Serial.print("");
  delay(200);
  Serial.write(0x0D);
  delay(200);
  Serial.println("Y=15Km");
  delay(200);

```

```

        Serial.write(0x1A);
        delay(200);
        j1 = 1;
        YY=1;
    }
}
else
if(YELLOWY>=510 && YELLOWY<=515 )
{

if(k1 ==0)
{
    lcd.setCursor(9, 0); // top left
    lcd.print("Y=10Km");
    digitalWrite (YELLOW2,LOW);
    Serial.print("AT\r");
    delay(200);
    Serial.print("AT+CMGF=1\r");
    delay(200);
    Serial.print("AT+CNMI=2,2,0,0,0\r");
    delay(200);
    Serial.print("AT+CMGS=");
    delay(200);
    Serial.print("");
    delay(200);
    Serial.print(num);
    delay(200);
    Serial.print("");
    delay(200);
    Serial.write(0x0D);
    delay(200);
    Serial.println("Y=10Km");
    delay(200);
    Serial.write(0x1A);
    delay(200);
    k1 = 1;
    YY=1;
}
}
else
if(YELLOWY<=400 )
{

if(l1 == 0)
{
    lcd.setCursor(9, 0); // top left
    lcd.print("Y=5Km");
    digitalWrite (YELLOW2,LOW);
    Serial.print("AT\r");
    delay(200);
    Serial.print("AT+CMGF=1\r");
    delay(200);

```

```

        Serial.print("AT+CNMI=2,2,0,0,0\r");
        delay(200);
        Serial.print("AT+CMGS=");
        delay(200);
        Serial.print("");
        delay(200);
        Serial.print(num);
        delay(200);
        Serial.print("");
        delay(200);
        Serial.write(0x0D);
        delay(200);
        Serial.println("Y=5Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);
    ll=1;
    YY=1;
}
}

}
}

void BLUE1()
{
    if(BB==0)
    {
        BLUEB = analogRead(analogInPin2);
        if(BLUEB>=850&& BLUEB<=856 )
        {
            if(m1 ==0)
            {
                lcd.setCursor(0, 1); // top left
                lcd.print("B=30Km");
                digitalWrite (BLUE2,LOW);
                Serial.print("AT\r");
                delay(200);
                Serial.print("AT+CMGF=1\r");
                delay(200);
                Serial.print("AT+CNMI=2,2,0,0,0\r");
                delay(200);
                Serial.print("AT+CMGS=");
                delay(200);
                Serial.print("");
                delay(200);
                Serial.print(num);
                delay(200);
                Serial.print("");
                delay(200);
                Serial.write(0x0D);
                delay(200);
            }
        }
    }
}

```

```

        Serial.println("B=30Km");
        delay(200);
        Serial.write(0x1A);
        delay(200);
    m1 =1;
    BB=1;
}
}
else if(BLUEB>=820 && BLUEB<=825 )
{

if(n1 ==0);
{
    lcd.setCursor(0, 1); // top left
    lcd.print("B=25Km");
    digitalWrite (BLUE2,LOW);
    Serial.print("AT\r");
    delay(200);
    Serial.print("AT+CMGF=1\r");
    delay(200);
    Serial.print("AT+CNMI=2,2,0,0,0\r");
    delay(200);
    Serial.print("AT+CMGS=");
    delay(200);
    Serial.print("");
    delay(200);
    Serial.print(num);
    delay(200);
    Serial.print("");
    delay(200);
    Serial.write(0x0D);
    delay(200);
    Serial.println("B=25Km");
    delay(200);
    Serial.write(0x1A);
    delay(200);
    n1 =1;
    BB=1;
}
}
else
if(BLUEB>=765 && BLUEB<=770 )
{
    if(o1 ==0)
    {

        lcd.setCursor(0, 1); // top left
        lcd.print("B=20Km");
        digitalWrite (BLUE2,LOW);
        Serial.print("AT\r");
        delay(200);
        Serial.print("AT+CMGF=1\r");

```

```

    delay(200);
    Serial.print("AT+CNMI=2,2,0,0,0\r");
    delay(200);
    Serial.print("AT+CMGS=");
    delay(200);
    Serial.print("");
    delay(200);
    Serial.print(num);
    delay(200);
    Serial.print("");
    delay(200);
    Serial.write(0x0D);
    delay(200);
    Serial.println("B=20Km");
    delay(200);
    Serial.write(0x1A);
    delay(200);
    o1 =1;
    BB=1;
  }
}
else
if(BLUEB>=670 && BLUEB<=695 )
{

if(p1 ==0)
{
  lcd.setCursor(0, 1); // top left
  lcd.print("B=15Km");
  digitalWrite (BLUE2,LOW);
  Serial.print("AT\r");
  delay(200);
  Serial.print("AT+CMGF=1\r");
  delay(200);
  Serial.print("AT+CNMI=2,2,0,0,0\r");
  delay(200);
  Serial.print("AT+CMGS=");
  delay(200);
  Serial.print("");
  delay(200);
  Serial.print(num);
  delay(200);
  Serial.print("");
  delay(200);
  Serial.write(0x0D);
  delay(200);
  Serial.println("B=15Km");
  delay(200);
  Serial.write(0x1A);
  delay(200);
  p1 =1;
  BB=1;

```

```

}
}
else
if(BLUEB>=510 && BLUEB<=515 )
{

if(q1 ==0)
{
  lcd.setCursor(0, 1); // top left
  lcd.print("B=10Km");
  digitalWrite (BLUE2,LOW);
  Serial.print("AT\r");
  delay(200);
  Serial.print("AT+CMGF=1\r");
  delay(200);
  Serial.print("AT+CNMI=2,2,0,0,0\r");
  delay(200);
  Serial.print("AT+CMGS=");
  delay(200);
  Serial.print("");
  delay(200);
  Serial.print(num);
  delay(200);
  Serial.print("");
  delay(200);
  Serial.write(0x0D);
  delay(200);
  Serial.println("B=10Km");
  delay(200);
  Serial.write(0x1A);
  delay(200);

  q1 =1;
  BB=1;
}
}
else
if(BLUEB<=400 )
{

if(r1 ==0)
{
  lcd.setCursor(0, 1); // top left
  lcd.print("B=5Km");
  digitalWrite (BLUE2,LOW);
  Serial.print("AT\r");
  delay(200);
  Serial.print("AT+CMGF=1\r");
  delay(200);
  Serial.print("AT+CNMI=2,2,0,0,0\r");
  delay(200);
  Serial.print("AT+CMGS=");
  delay(200);

```



```
    Serial.print("");  
    delay(200);  
    Serial.print(num);  
    delay(200);  
    Serial.print("");  
    delay(200);  
    Serial.write(0x0D);  
    delay(200);  
    Serial.println("B=5Km");  
    delay(200);  
    Serial.write(0x1A);  
    delay(200);  
    r1 =1;  
    BB=1;  
    }  
    }  
    }  
}  
}
```

CHAPTER 6

RESULT & APPLICATIONS

6.1 RESULT



Figure 6.1 : Model Output

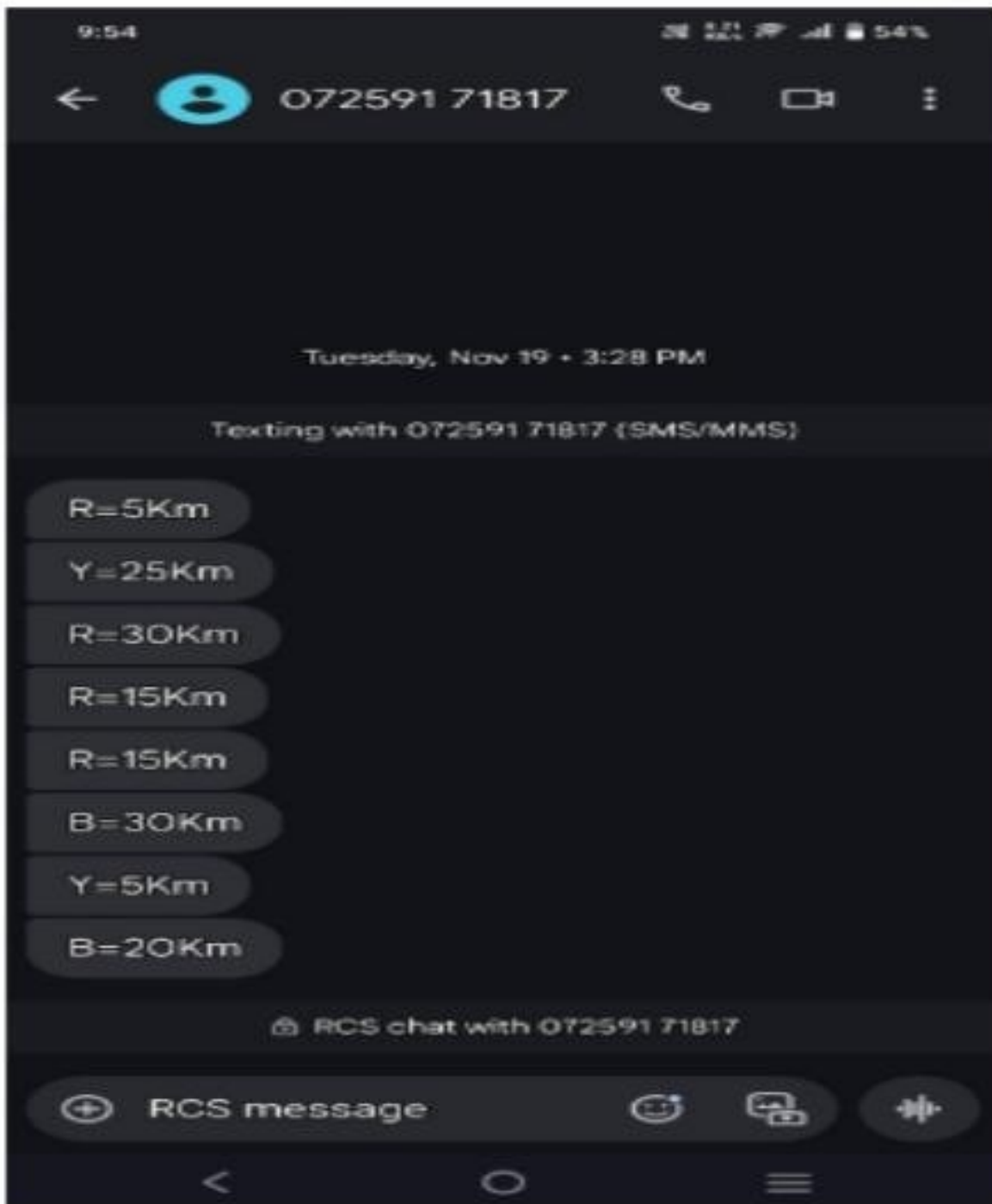


Figure 6.2 SMS of output

Explanation of the project:

- 1) The cable is divided into **equal segments** using resistors to simulate length.
- 2) The Arduino continuously monitors voltage at each segment.
- 3) When a fault (short or open) is simulated, the voltage at a certain segment drops.
- 4) Arduino calculates the distance of the fault using the formula:
- 5) **Distance = Measured Voltage / Total Voltage × Cable Length**

$$\text{Distance} = \frac{\text{Measured Voltage}}{\text{Total Voltage}} \times \text{Cable Length}$$
- 6) The location of the fault is displayed on the LCD.
- 7) The **GSM module** sends an SMS with the **fault location** to the user.

Basic Principle – Ohm's Law

Ohm's Law:

$$V = I \times R \quad V = I \times R \quad V = I \times R$$

Where: **V = Voltage**

I = Current

R = Resistance

Resistors simulate the cable. A fault (like a short circuit) causes a **drop in voltage** along the cable. By measuring the voltage drop, we can estimate **how far the fault is** from the source.

Project Setup

1. We have **6 resistors**, each representing **5 km** of cable:
2. **Total Length = 6 × 5 = 30 km**

$$\text{Total Length} = 6 \times 5 = 30 \text{ km}$$
3. **Each resistor = 1 segment = 5 km**
4. So, resistance increases linearly with cable length.

Mathematical Fault Detection Logic

Total resistance = 6Ω (1Ω per 5 km)

Supply voltage = 5V

We use a **voltage divider rule** to detect at which point the voltage drops significantly.

Voltage Divider Formula:

$$V_{\text{out}} = V_{\text{in}} \times \frac{R_{\text{measured}}}{R_{\text{total}}} \quad V_{\text{out}} = V_{\text{in}} \times \frac{R_{\text{measured}}}{R_{\text{total}}}$$

If fault occurs after 2 resistors (10 km), the voltage will only appear across those 2 resistors:

$$V = 5V \times \frac{2\Omega}{6\Omega} = 1.67V \quad V = 5V \times \frac{2\Omega}{6\Omega} = \frac{10}{6} = 1.67V$$

Arduino reads 1.67V → this tells us fault is at 2nd segment (i.e., 10 km)

Summary of Steps

- 1) Supply 5V across 6 resistors (cable segments)
- 2) Arduino reads voltage at each segment
- 3) When voltage drops at a segment, it calculates: $\text{Fault Distance} = \frac{V}{V_{\text{total}}} \times 30 \text{ km}$
- 4) Displays distance on LCD
- 5) Sends SMS via GSM with the same message

Distance Calculation

If fault is detected at segment i, distance is:

- $\text{Distance} = (i+1 \text{ Total segments}) \times \text{Total cable length}$

- For example: If fault is found at A1 ($i = 1$), $\text{Distance} = (26) \times 30 = 10 \text{ km}$
 $\left(\frac{2}{6}\right) \times 30 = 10 \text{ km}$
 $\text{Distance} = (62) \times 30 = 10 \text{ km}$

6.2 APPLICATIONS

- 1) Utility and Power Line Maintenance
- 2) Telecommunication Cable Detection
- 3) Pipeline and Gas Line Detection
- 4) Railway Track Maintenance
- 5) Municipal Water and Sewage Systems
- 6) Mapping and Surveying

6.3 ADVANTAGES

- 1) Less Maintenance.
- 2) Less complexity.
- 3) It has higher Efficiency.
- 4) It can detect other types of cable fault such as Short circuit fault, cable cuts.
- 5) Resistive fault, Sheath faults, Water trees, Partial discharges.
- 6) Consume low power in Nano watts.
- 7) Provide precise accuracy in determining the Location of Fault.

6.4 DISADVANTAGES

- 1) System failure can damage the components.
- 2) No Network it will not work.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

7.1 CONCLUSION

Many times, during construction or excavation projects, underground cables are accidentally damaged or disturbed, leading to potential hazards and service interruptions. In such cases, it can be challenging to locate and identify the cables in the ground. The main proposal for this project is to enhance safety and reduce risks by providing a reliable underground cable detection system. This idea will minimize overall costs through the use of low-cost sensors, microcontrollers like Arduino, and wireless communication technologies.

Overall, the underground cable detection system offers an efficient and secure way to detect the presence and location of underground cables, reducing the chances of accidental damage during digging or construction work. The system's effectiveness relies on the accuracy of the sensors and the reliability of the communication technologies used. Furthermore, implementing real-time monitoring can provide immediate alerts if cables are accidentally damaged or exposed, enhancing overall safety during excavation activities.

The conclusion of the discussion on the underground cable detection system is that it represents an innovative approach to managing underground infrastructure. By utilizing Arduino-based technology, the system is cost-effective and simple to implement. This system's design simplicity allows for quick installation and makes it a viable solution for a variety of applications, from construction sites to maintenance projects. In conclusion, the project meets the intended specifications and can be effectively deployed to ensure safer and more efficient underground cable management. The use of Arduino and sensor technology in this project ensures that the detection system is not only affordable but also highly scalable for future expansions and improvements.

7.2 FUTURE SCOPE

The future scope for this underground cable detection system includes further enhancements that improve accuracy, reliability, and ease of use. One potential extension is integrating the system with a cloud-based platform, enabling real-time monitoring and data storage. This will allow users to track cable locations and maintenance records remotely via mobile or desktop applications. Additionally, by incorporating GPS technology, the system can provide precise location data of detected cables, which will be particularly useful for large-scale infrastructure projects.

Another future feature involves integrating AI-powered analytics to predict potential faults or wear in the cables based on historical data and environmental conditions. This predictive maintenance approach will help prevent disruptions and optimize repair schedules. Furthermore, a mobile app extension could alert users instantly if cables are detected as damaged, allowing for immediate action to be taken before the issue escalates.

To increase safety, the system could include a real-time alert feature, notifying the user if any unauthorized digging or construction is detected near vital underground cables. The app can send instant push notifications and location data of the potential threat. Finally, integrating sensors capable of detecting changes in the underground environment (e.g., moisture or pressure) could allow for proactive cable management, ensuring more secure and long-lasting underground infrastructure.

APPENDIX B

Evolution Sheet (ESE)

For

Capstone Project Planning

Name of Students: Dhruv Ahire, Rohan Awate, Nirbhay Chalke,

Pratik Gharwade

Name of Programme: Electronic & Tele-communication Engineering.

Semester: 6th

Course Title: Capstone Project Planning.

Code:22058

Title of Capstone Project: Underground Cable Fault Detection

A. POS addressed by the Capstone Project POS).

PO1: Basic and discipline-specific knowledge: Apply knowledge of basic mathematics, science, and engineering specialization to solve the engineering problems.

PO2: Problem analysis: Identify and analyze well-defined engineering problems using codified standard methods.

PO3: Design/development of solutions: Design solutions for well-defined technical problems and assist with the design of systems components or processes to meet specified needs.

PO4: Engineering Tools, experimentation, and Testing: Apply modern engineering tools and appropriate techniques to conduct standard tests and measurements.

PO5: Engineering practices for society, sustainability, and environment: Apply appropriate technology in the context of society, sustainability, environment, and ethical practices.

PO6: Project management: Use engineering management principles individually, as a team member, or as a leader to manage the project and effectively communicate about well-defined engineering activities.

PO7: Life-long learning: Ability to analyze individual needs and engage in updating in the context of technological changes.

B. Program Specific Outcomes (PSO) addressed by the capstone project.

PSO1. Electronics and Telecommunication System: Maintain various types of Electronics and Telecommunication engineering specific industry soon after the diploma program.

PSO2. EDA Tools Usage: Use EDA Tools to develop simple Electronics and Telecommunications Engineering related Circuits.

C. COS addressed by the Capstone Project POS).

- a) Implement the planned activity individually and/or as a team.
- b) Select, collect, and use required information/knowledge to solve the identified problem
- c) Take appropriate decisions based on collected and analyzed information.
- d) Ensure the quality of the product.
- e) Consider the ethical issues related to the project.
- f) Assess the impact of the project on society.
- g) Communicate effectively and confidently as a member and leader of the team.
- h) Prepare project report after performing due plagiarism check using appropriate tools.

I. Other learning outcomes achieved through this project

1. Unit Outcomes (Cognitive Domain)

- a) Tested the Performance of Network Devices (GSM 800A Module)
- b) Interpret of Execution of code using relevant software.
- c) Develop the Flowchart on the basis of Coding.
- d) Troubleshooting each Electronic component used in the project using relevant tools (DMM).

2. Practical Outcomes (in Psychomotor Domain)

- a) Interpret the working of Components using an Open-source electronics platform (Arduino).
- b) Maintain a control circuit consisting of electronic devices.
- c) Interpret the working of the underground cable fault detection using an Open-source Platform (Arduino).
- d) Interpret and Execute the Code using an open-source electronics platform (Arduino)

3. Affective Domain Outcomes

- a) Follow safe Practices.
- b) Demonstrate Working as a leader / a team member.
- c) Maintain Electronics Equipment.
- d) Perform under Guidance.

PROGRESSIVE ASSESSMENT (PA) SHEET		
Sr No.	Criteria	Marks
1	Project Proposal / Identification	10
2	Punctuality and Overall Contribution	
3	Project Diary	
4	Execution of Plan During the Sixth Semester	20
5	Project Report Including Documentation	15
6	Presentation	05
Total		50

PROGRESSIVEASSESSMENT SHEET (PA)								
Name of Student	Roll No	Project Proposal / identification (2)	Punctuality and Overall Contribution (2)	Project Diary (6)	Industrial Survey / Literature Review (20)	Report writing (15)	Presentation (05)	Total (50)
Dhruv Ahire	03							
Rohan Awate	05							
Nirbhay Chalke	11							
Pratik Gharwade	18							

Name and designation of the faculty member: Mrs. Vaishali Rajeshirke

Signature:- _____

APPENDIX

SUGGESTED RUBRIC FOR ASSESSMENT OF CAPSTONE PROJECT

Sr No.	Characteristic to be assessed	Poor Marks (1 - 3)	Average marks (1 - 3)	Good Marks (1 - 3)	Excellent Marks (1 - 3)
1.	Problem/ Task Identification (Project Title)	Relates to very few POs Scope of problem not clear at all	Related to Some POs Scope of Problem / task vague	Take care of at least there POs Scope of Problem / Task not very specified	Take care of more than Three POs Scope of Problem/ Task very clear
2.	Literature Survey/ IndustrialSurvey	Not more than Ten sources (Primary and secondary) ,very old reference	At-least 10 relevant sources, most latest	At-least 15 relevant sources, most latest	About 20 relevant sources, most latest
3.	Project Proposal	Methods are not appropriate, All Steps not Mention Design of the Prototype not started (if applicable)	Appropriate plan but not in much detail. Plan B for critical activities not mentioned. Timeline is not developed. Design of Prototype is Not complete. (if applicable)	Appropriate and detailed plan With Plan B for Critical activities mentioned, but clarity is not There in methods, time The line is given but not appropriate Design of Prototype isnot detailed	Appropriate and detailed plan with Plan B for critical activities mentioned, clarity in methods with timeline, detail design of prototype(if applicable)
4.	Project Diary	Entries for most	Entries for some	Entries were	Entries were

		weeks are missing. There is no proper sequence and details are not correct.	weeks are missing. Details are not appropriate, not signed regularly by the guide.	made every Week but are not in detail. Signed And approved by A guide every week	Made every week in detail, signed and approved by Guide every week
--	--	---	--	--	--

5.	Final report Presentation	Very Short, Poor quality sketches, detail about methods, material, precautions and conclusion omitted, some details are wrong	Detail, correct and clear descriptions of method, Materials and precautions	Conclusion, sufficient graphic description	Very Detail, correct, clear description of methods, materials, precautions and conclusions. Enough tables, charts and sketches
6.	Presentation	Major information is not including, Information is Not well organized	Include information but not well organized and Not presented well	Includes major information and well organized but not Presented well	Well organized, include major information, Well present
7.	Question and Answer session	Could not reply To considerable Number of question	Replies to considerable Number of Question but not very properly	Reply properly to considerable Number of Questions	Reply to most Of the Question properly

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